Measurement Error Models

General Principles

Measurement error refers to the variability in the measurement of a variable, and measurement error can be generated by several factors, such as sampling bias, censoring bias, and group size heterogeneity. It is an important consideration in many fields, including statistics, economics, and engineering, where accurate measurements are crucial for making informed decisions. To account for measurement error, we can use a *measurement error model*. This model assumes that the measurement of a variable is subject to an error, which can be modeled using a probability distribution. The model can be used to estimate the parameters of the measurement error distribution, such as the mean and variance, and to make predictions about the measurements based on the estimated parameters. Measurement error models are *composed models* (i.e., models with sub-models) that evaluate different generative processes, starting with the measurement error process, which is then used to generate the observed data.

Example

Below is an example code snippet demonstrating a Bayesian measurement error model using the BayesForge (BF) package. The data consist of three continuous variables (marriage rate, divorce rate, age), and the goal is to estimate the effect of age and marriage rate on the divorce rate while considering that the divorce rate has a measurement error. This example is based on McElreath (2018).

Python

```python
from BayesForge import bf
import jax.numpy as jnp

# Setup device-----------------------------------------------
m = bf(platform='cpu')
```

```
# Import Data & Data Manipulation ------------------------------------------------
# Import
data_path = m.load.WaffleDivorce(only_path=True)
m.data(data_path, sep=';')
m.scale(['MedianAgeMarriage', 'Marriage']) # Scale
dat = dict(
    D_obs = m.z_score(m.df['Divorce'].values),
    D_sd = jnp.array(m.df['Divorce SE'].values / m.df['Divorce'].std()),
    A = jnp.array(m.df['MedianAgeMarriage'].values),
    M = jnp.array(m.df['Marriage'].values),
    N = m.df.shape[0]
)
m.data_on_model = dat # Send to model (convert to jax array)

# Define model -------------------------------------------------
def model(D_obs, D_sd, A, N, M):
    a = m.dist.normal(0, 0.2, name = 'a')
    beta = m.dist.normal(0, 0.5, name = 'beta')
    eta = m.dist.normal(0, 0.5, name = 'eta')
    s = m.dist.exponential(1, name = 's')
    mu = a + beta * A + eta * M
    D_true = m.dist.normal(mu, s, name = 'D_true')
    m.dist.normal(D_true , D_sd, obs = D_obs)

# Run MCMC -----------------------------------------------------
m.fit(model, progress_bar=False)  # Optimize model parameters through MCMC sampling

# Summary ------------------------------------------------------
m.summary() # Get posterior distributions
```

```
An NVIDIA GPU may be present on this machine, but a CUDA-enabled jaxlib is not installed. Fal

bf v 0.0.60 package loaded
jax.local_device_count 16
```

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	mcse_mean	mcse_sd
D_true[0]	1.19	0.37	0.61	1.79	5831.69	2891.03	1.00	0.00	0.01
D_true[1]	0.71	0.57	-0.20	1.63	7461.94	3093.72	1.00	0.01	0.01
D_true[2]	0.44	0.34	-0.11	1.00	8515.85	3218.57	1.00	0.00	0.01
D_true[3]	1.44	0.47	0.70	2.19	6394.72	2783.68	1.00	0.01	0.01

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	mcse_mean	mcse_sd
D_true[4]	-0.91	0.13	-1.12	-0.70	10060.24	2896.50	1.00	0.00	0.00
D_true[5]	0.68	0.40	0.04	1.33	7931.92	2456.83	1.00	0.00	0.01
D_true[6]	-1.40	0.36	-1.97	-0.83	8511.16	2982.53	1.00	0.00	0.01
D_true[7]	-0.33	0.49	-1.12	0.45	7847.42	3219.95	1.00	0.01	0.01
D_true[8]	-1.89	0.59	-2.82	-0.94	5974.10	3294.37	1.00	0.01	0.01
D_true[9]	-0.62	0.17	-0.89	-0.35	9345.29	2652.40	1.00	0.00	0.00
D_true[10]	0.79	0.28	0.34	1.24	7133.14	3004.76	1.00	0.00	0.01
D_true[11]	-0.56	0.49	-1.33	0.20	6190.13	3304.74	1.00	0.01	0.01
D_true[12]	0.15	0.50	-0.67	0.93	3698.49	3099.26	1.00	0.01	0.01
D_true[13]	-0.88	0.23	-1.25	-0.51	8777.07	2876.02	1.00	0.00	0.00
D_true[14]	0.57	0.30	0.09	1.05	7332.69	2746.10	1.00	0.00	0.01
D_true[15]	0.29	0.37	-0.33	0.89	11138.20	2980.48	1.00	0.00	0.01
D_true[16]	0.50	0.43	-0.19	1.18	8481.19	2885.68	1.00	0.00	0.01
D_true[17]	1.28	0.36	0.72	1.86	8792.99	3155.58	1.00	0.00	0.01
D_true[18]	0.45	0.38	-0.15	1.06	9732.54	3154.51	1.00	0.00	0.01
D_true[19]	0.43	0.55	-0.42	1.32	4467.29	3169.51	1.00	0.01	0.01
D_true[20]	-0.55	0.32	-1.05	-0.05	9506.91	2717.05	1.00	0.00	0.01
D_true[21]	-1.11	0.26	-1.52	-0.68	9180.62	3104.37	1.00	0.00	0.00
D_true[22]	-0.28	0.26	-0.70	0.14	8758.82	2735.76	1.00	0.00	0.00
D_true[23]	-1.02	0.30	-1.50	-0.53	7374.03	2824.29	1.00	0.00	0.01
D_true[24]	0.44	0.40	-0.20	1.09	7912.80	2931.77	1.00	0.00	0.01
D_true[25]	-0.03	0.32	-0.54	0.47	11730.29	3410.83	1.00	0.00	0.01
D_true[26]	-0.03	0.51	-0.85	0.78	6993.33	2863.17	1.00	0.01	0.01
D_true[27]	-0.16	0.40	-0.80	0.46	9576.22	2903.85	1.00	0.00	0.01
D_true[28]	-0.26	0.52	-1.05	0.59	7447.36	2859.77	1.00	0.01	0.01
D_true[29]	-1.82	0.23	-2.19	-1.44	8544.76	2967.29	1.00	0.00	0.00
D_true[30]	0.18	0.43	-0.51	0.88	10136.25	2921.82	1.00	0.00	0.01
D_true[31]	-1.68	0.17	-1.95	-1.41	10853.79	2962.74	1.00	0.00	0.00
D_true[32]	0.12	0.24	-0.26	0.50	10320.18	2901.63	1.00	0.00	0.00
D_true[33]	-0.08	0.52	-0.91	0.73	5111.73	2850.62	1.00	0.01	0.01
D_true[34]	-0.12	0.23	-0.49	0.25	8954.77	2803.13	1.00	0.00	0.00
D_true[35]	1.30	0.42	0.64	1.98	7124.41	3321.79	1.00	0.00	0.01
D_true[36]	0.24	0.36	-0.33	0.82	9735.32	2653.16	1.00	0.00	0.01
D_true[37]	-1.03	0.22	-1.38	-0.68	9791.92	3088.44	1.01	0.00	0.00
D_true[38]	-0.91	0.53	-1.74	-0.05	6296.40	3433.55	1.00	0.01	0.01
D_true[39]	-0.69	0.32	-1.22	-0.18	8436.51	2900.86	1.00	0.00	0.01
D_true[40]	0.26	0.56	-0.63	1.16	9294.63	2910.39	1.00	0.01	0.01
D_true[41]	0.75	0.35	0.19	1.31	8177.82	2743.01	1.00	0.00	0.01
D_true[42]	0.20	0.18	-0.09	0.48	9556.25	2844.95	1.00	0.00	0.00
D_true[43]	0.80	0.43	0.10	1.46	4869.33	3307.01	1.00	0.01	0.01
D_true[44]	-0.40	0.53	-1.23	0.45	7814.38	3192.47	1.00	0.01	0.01

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	mcse_mean	mcse_sd
D_true[45]	-0.40	0.26	-0.81	0.01	10237.21	3034.59	1.00	0.00	0.00
D_true[46]	0.14	0.31	-0.35	0.63	10299.71	3059.56	1.00	0.00	0.01
D_true[47]	0.57	0.47	-0.15	1.32	8946.69	2930.81	1.00	0.00	0.01
D_true[48]	-0.65	0.28	-1.09	-0.21	9877.18	3103.90	1.00	0.00	0.01
D_true[49]	0.85	0.59	-0.11	1.78	5627.28	3085.92	1.00	0.01	0.01
a	-0.05	0.10	-0.20	0.10	5242.71	3811.29	1.00	0.00	0.00
beta	-0.61	0.16	-0.87	-0.37	3922.34	3533.22	1.00	0.00	0.00
eta	0.06	0.17	-0.22	0.33	3622.32	3098.57	1.00	0.00	0.00
s	0.60	0.11	0.43	0.78	1317.80	1715.96	1.00	0.00	0.00

R

```
library(BayesForge)
jnp = reticulate::import('jax.numpy')

# Setup platform-----------------------------------------------
m=importBF(platform='cpu')

# Import data -------------------------------------------------
m$data(m$load$WaffleDivorce(only_path = TRUE), sep=';')

m$scale(list('MedianAgeMarriage', 'Marriage'))

m$data_on_model$D_obs = m$z_score(jnp$array(m$df['Divorce']))
m$data_on_model$D_sd = jnp$array(m$df['Divorce SE']) / sd(unlist(m$df['Divorce']))
m$data_on_model$A = jnp$array(m$df['MedianAgeMarriage'])
m$data_on_model$M = jnp$array(m$df['Marriage'])
m$data_on_model$N = as.integer(nrow(m$df))

# Define model ------------------------------------------------
model <- function(D_obs, D_sd, A, N, M){
  a = m$dist$normal(0, 0.2, name = 'a')
  beta = m$dist$normal(0, 0.5, name = 'beta')
  eta = m$dist$normal(0, 0.5, name = 'eta')
  s = m$dist$exponential(1, name = 's')
  mu = a + beta * A + eta * M
  D_true = m$dist$normal(mu, s, name = 'D_true')
  m$dist$normal(D_true , D_sd, obs = D_obs, name = 'D_obs')
}
```

```
# Run MCMC -----------------------------------------------
m$fit(model) # Optimize model parameters through MCMC sampling

# Summary ------------------------------------------------
m$summary() # Get posterior distribution
```

Julia

```
using PyBayesForge

# Setup device-------------------------------------------------
m = importBF(platform="cpu")

# Import Data & Data Manipulation ------------------------------------------------
# Import
data_path = m.load.WaffleDivorce(only_path=true)
m.data(data_path, sep=";")
m.scale(["MedianAgeMarriage", "Marriage"]) # Scale
dat = pydict(
    D_obs = m.z_score(m.df["Divorce"].values),
    D_sd = jnp.array(m.df["Divorce SE"].values / m.df["Divorce"].std()),
    A = jnp.array(m.df["MedianAgeMarriage"].values),
    M = jnp.array(m.df["Marriage"].values),
    N = m.df.shape[0]
)
m.data_on_model = dat # Send to model (convert to jax array)

# Define model -----------------------------------------------
@BF function model(D_obs, D_sd, A, N, M)
    a = m.dist.normal(0, 0.2, name = "a")
    beta = m.dist.normal(0, 0.5, name = "beta")
    eta = m.dist.normal(0, 0.5, name = "eta")
    s = m.dist.exponential(1, name = "s")
    mu = a + beta * A + eta * M
    D_true = m.dist.normal(mu, s, name = "D_true")
    m.dist.normal(D_true , D_sd, obs = D_obs)


end
```

```
# Run mcmc -----------------------------------------------
m.fit(model)  # Optimize model parameters through MCMC sampling

# Summary ------------------------------------------------
m.summary() # Get posterior distributions
```

Mathematical Details

Bayesian formulation

$$D_i^* \sim \text{Normal}(D_i, \varsigma_i)$$

$$D_i \sim \text{Normal}(\mu_i, \sigma)$$

$$\mu_i = \alpha + \beta A_i + \eta M_i$$

$$\sigma \sim \text{Normal}(1)$$

where:

- D_i^* is the observed divorce rate.
- D_i is the true divorce rate.
- μ_i is the mean of the true divorce rate.
- σ is the standard deviation of the true divorce rate.
- α is the intercept term.
- β is the regression coefficient for age.
- η is the regression coefficient for marriage rate.

Model validation tests

- Comparison with Stan: `15.Measuring error.py` fits the same model with BayesForge and Stan and compares the posterior distributions (KL divergence) and parameter recovery.
- Documentation examples: the Python, R and Julia examples of this page are run by the documentation tests (`Python/test_model_pages.py`, `R/test_r_model_pages.py`, `Julia/test_julia_model_pages.py`).

Notes

> **Note**
>
> This is an approach that can be extended to any kind of model previously described. For example, one could generate a Bernoulli measurement error model by generating a process for the probabilities of success and failure. We can even go further by potentially having an error rate that is present only in one of the two outcomes.

Reference(s)

McElreath, Richard. 2018. *Statistical Rethinking: A Bayesian course with examples in R and Stan*. Chapman; Hall/CRC.